

29	At time t, a sample of a radioactive substance contains N atoms of a particular nuclide. At time $(t + \Delta t)$, where Δt is a short period of time, the number of atoms of the nuclide is $(N - \Delta N)$. Which expression is equal to the decay constant of the nuclide?							
	A	$\frac{N\Delta N}{\Delta t}$	B	$\frac{\Delta N}{N\Delta t}$	C	$\frac{\Delta N}{\Delta t}$	D	$\frac{\Delta N}{N}$

Answer: B

Activity A = λN

For a short time interval,

$$\text{Activity } A = \frac{\Delta N}{\Delta t}$$

Hence

$$\frac{\Delta N}{\Delta t} = \lambda N$$

$$\lambda = \frac{\Delta N}{N \Delta t}$$

30	At time $t = 0$ some radioactive gas is injected into a cooled vessel. At time T a different
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